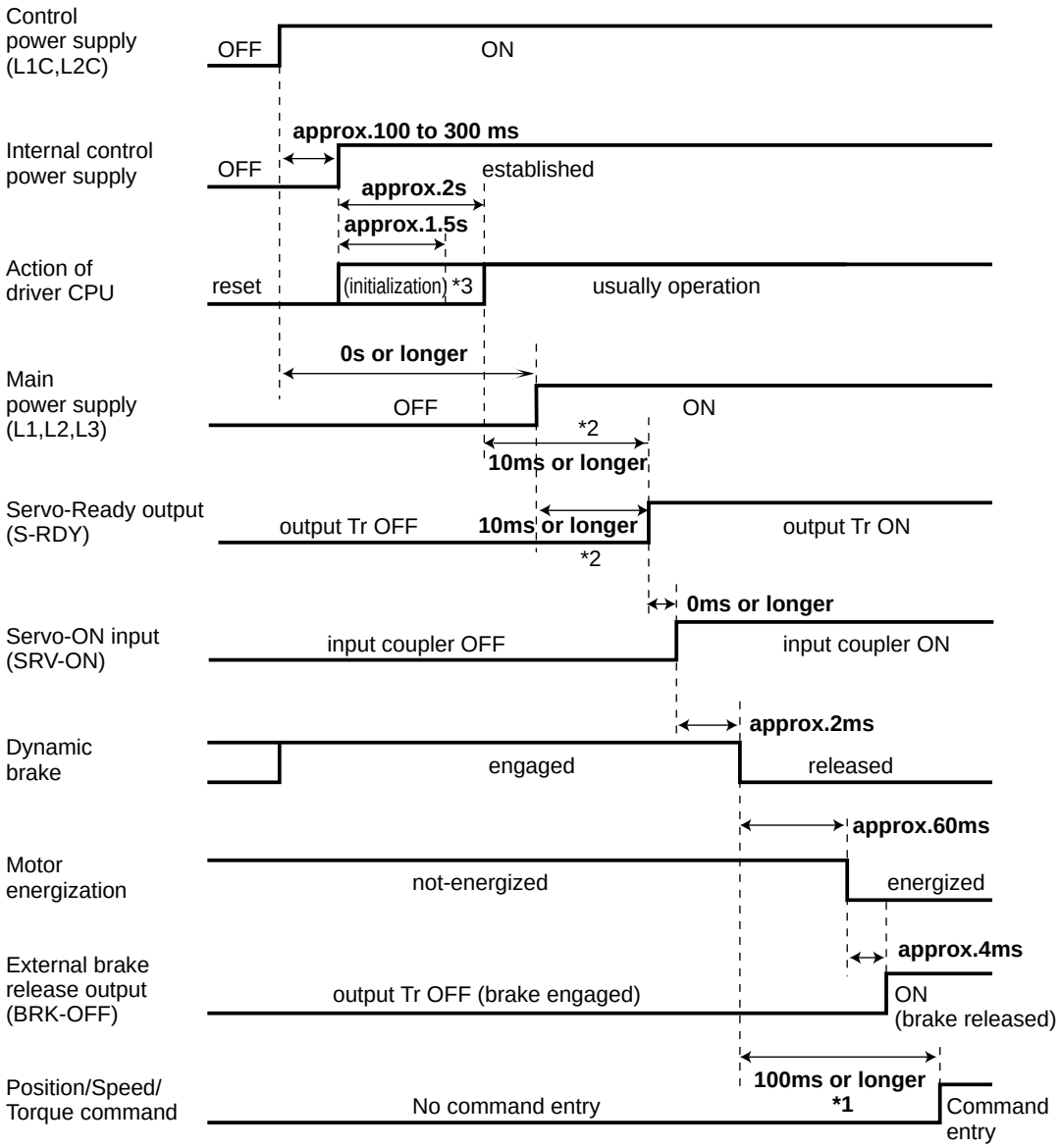
10. Timing Chart

Timing on power-up

Servo-on signal accept timing on power-up



- The above chart shows the timing from AC power-ON to command input.
- Activate the external command input according to the above timing chart.

Caution

*1. In this term Servo-ON input (SRV-ON) turns ON as a hard ware, but operation command can not be received.

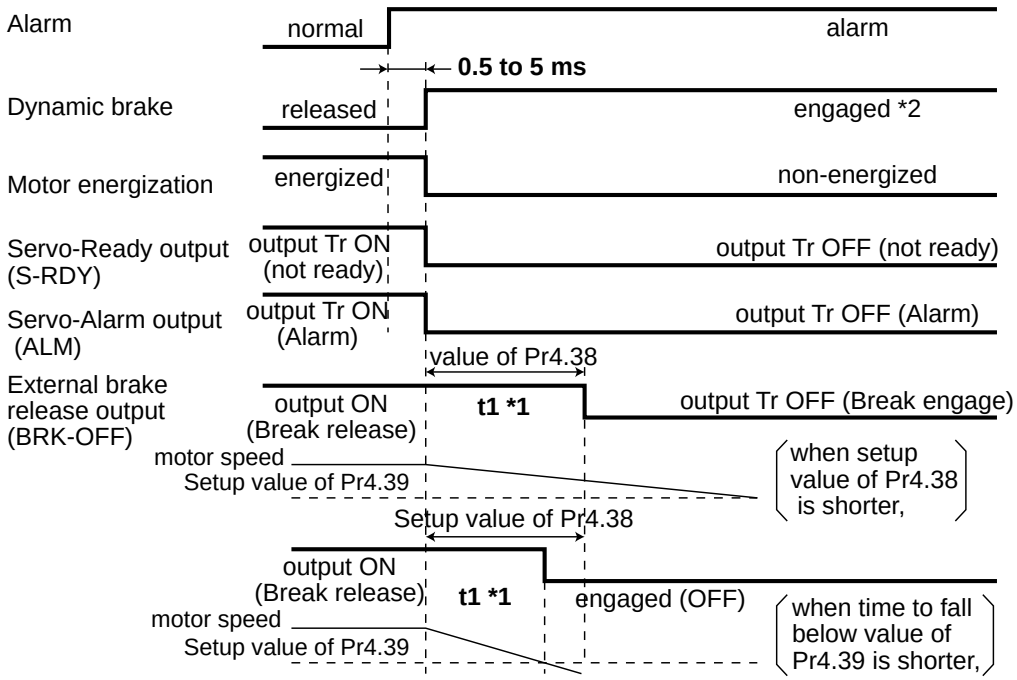
*2. S-RDY output will turn on when both conditions are met, initialization of micro computer has been completed and the main power has been turned on.

*3. After Internal control power supply , protective functions are active from approx. 1.5 sec after the start of initializing microcomputer. Please set the signals, especially for protective function, for example over-travel inhibit input (POT, NOT) or external scale input, so as to decide their logic until this term.
The lapse time can be changed with Pr6.18 Wait time after power-up.

10. Timing Chart

Alarm

When an Error (Alarm) Has Occurred (at Servo-ON Command)

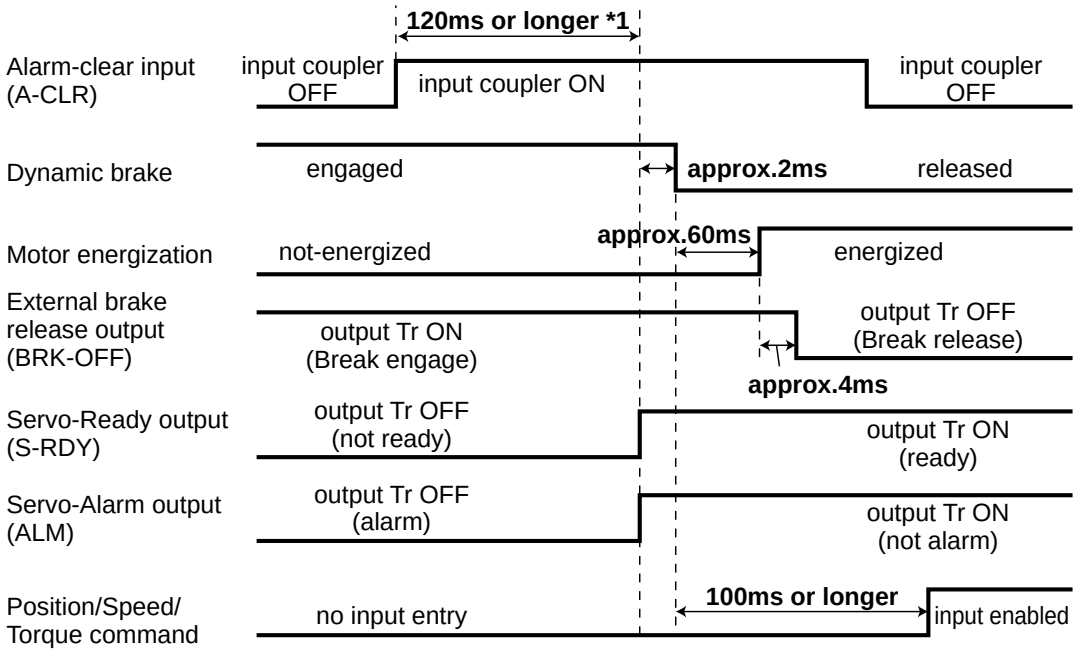


Caution

*1. t1 will be a shorter time of either the setup value of Pr4.38[Mechanical brake action at running setup] or elapsing time for the motor speed to fall below Pr4.39[Brake release speed setup].
t1 will be 0 when the motor is in stall regardless of the setup pf Pr4.37.

*2. When an alarm is generated, the dynamic brake operates according to Pr5.10 Sequence at alarm.

When an Alarm Has Been Cleared (at Servo-ON Command)



Caution

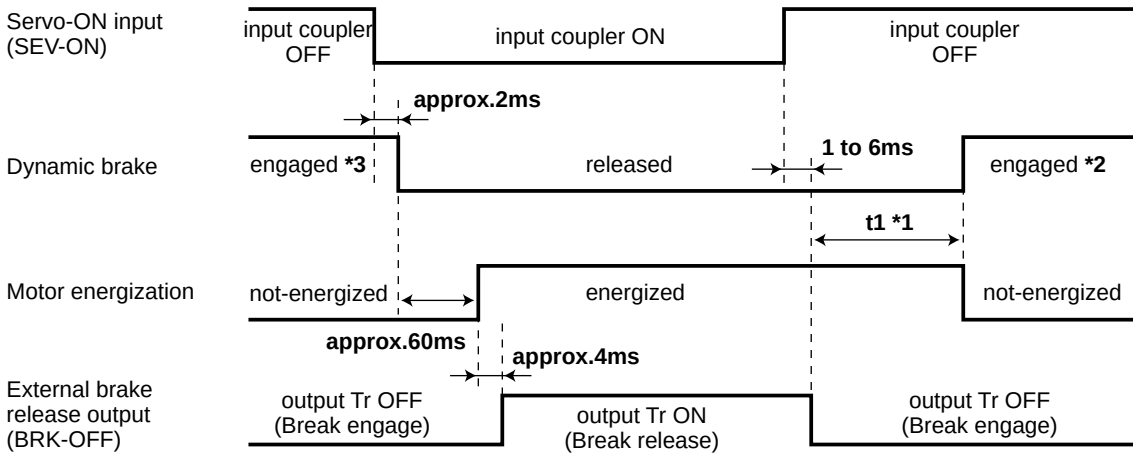
*1. The alarm clear input recognition time can be changed in Pr5.16 Alarm clear input setup.

10. Timing Chart

Servo-Lock

Servo-ON/OFF Action While the Motor Is at Stall (Servo-Lock)

Remarks **To turn on/off the servo during normal operation, first stop the motor.**



Caution

*1 t1 depends on the setup value of Pr4.37 Setup of mechanical brake action at stalling.

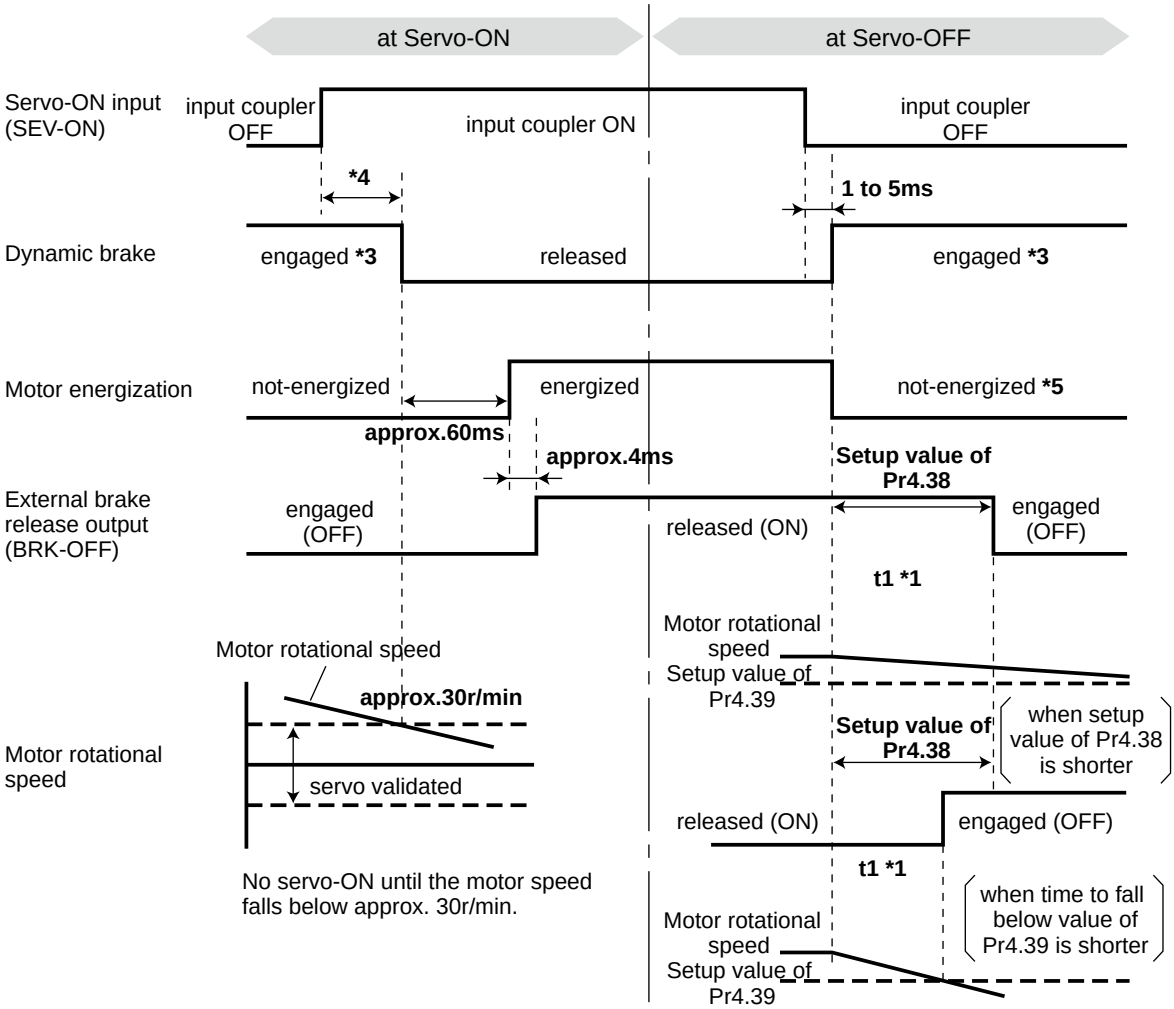
*2 The operation of dynamic brake during servo off depends on the setup value of Pr5.06 Sequence at servo off.

*3. Servo-ON will not be activated until the motor speed falls below approx. 30r/min.

Related page • P.4-40 “Details of Parameter”

Servo-ON/OFF Action While the Motor Is in Motion

Remarks **Timing at emergency stop or trip. Do not repeat this sequence.**



Caution

*1. t1 will be a shorter time of either the setup value of Pr4.38 “Mechanical brake action at running setup” or elapsing time for the motor speed to fall below Pr4.39 “Brake release speed setup”.

*2. Even though the SRV-ON signal is turned on again during the motor deceleration, Servo-ON will not be activated until the motor stops.

*3. For the action of dynamic brake at alarm occurrence, refer to an explanation of Pr5.06, “Sequence at Servo-OFF” as well.

*4. Servo-ON will not be activated until the motor speed falls below approx. 30r/min.

*5. For the motor energization during deceleration at Servo-OFF depends on the setup value of Pr.5.08, “Sequence at Servo-OFF”.

Related page • P.2-34 “Dynamic brake”